



PREDICTING INCIDENTAL DANGER



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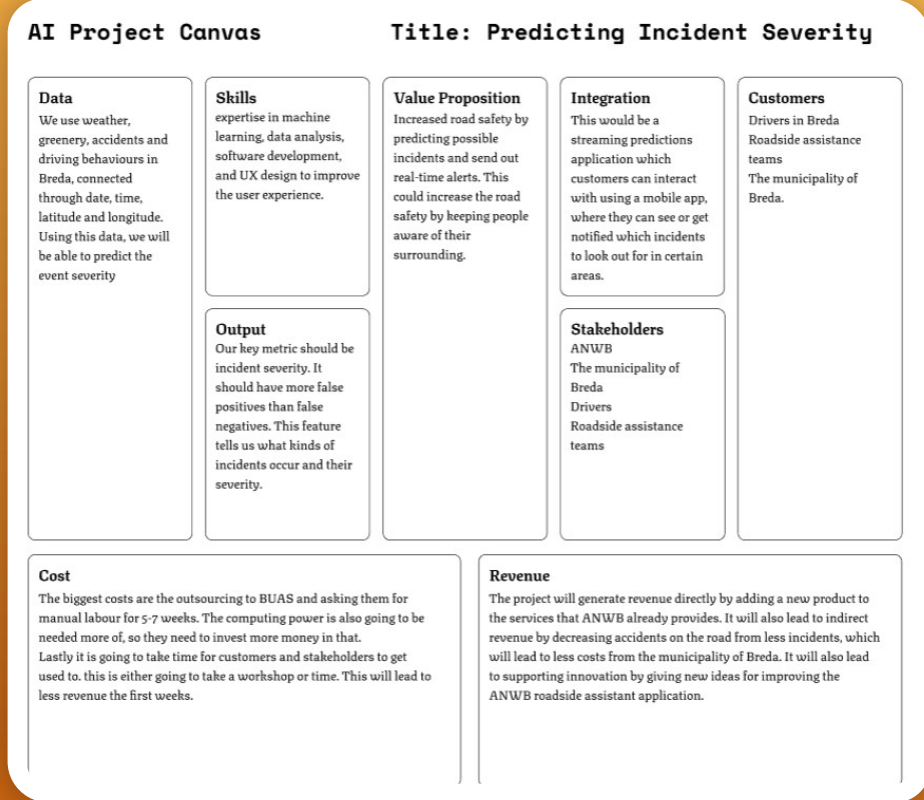


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Introduction

- Partnered with ANWB and municipality Breda
- Identify Reckless Driving Behaviours



Business Value

- What is our Business Value?
- Customers satisfaction
- Cost-Effective
- Road-Improvement
- Road-safety



Data Pre-Processing

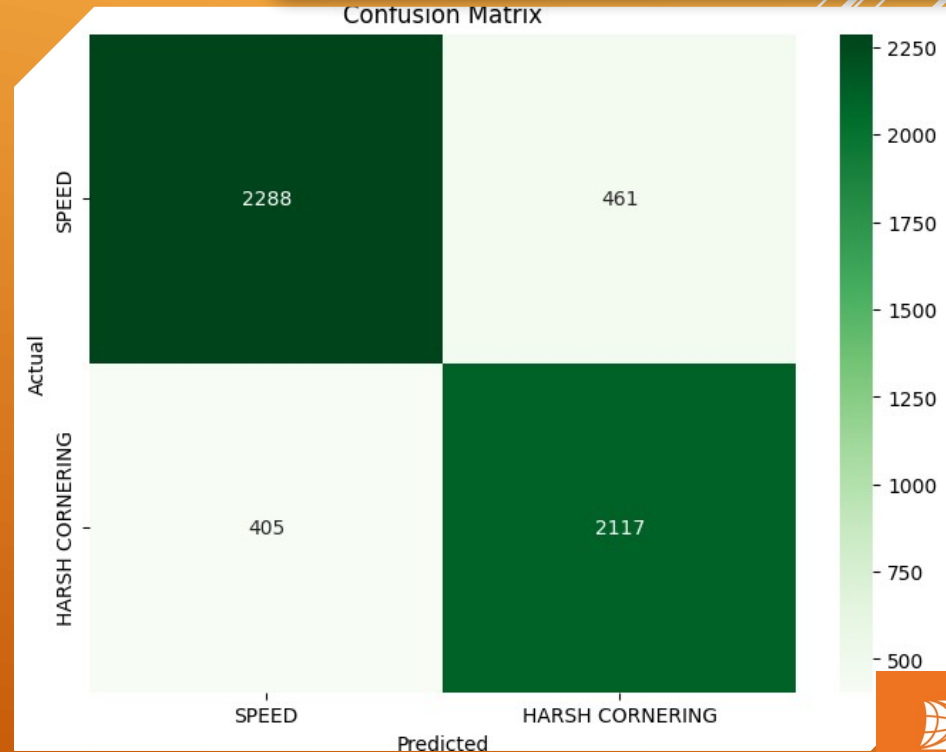
- ▮ Data Selection
- ▮ View Creation
- ▮ Filling Missing Values
- ▮ Handling date columns
- ▮ Categorical to Numerical
- ▮ Normalization
- ▮ Distribution Analysis
- ▮ Merging datasets
- ▮ Clustering Locations



Machine Learning Model

```
Training Accuracy: 0.8625978338208554
Test Accuracy: 0.8357047998482261
Precision: 0.8211792086889061
Recall: 0.8394131641554322
F1 Score: 0.8301960784313726
```

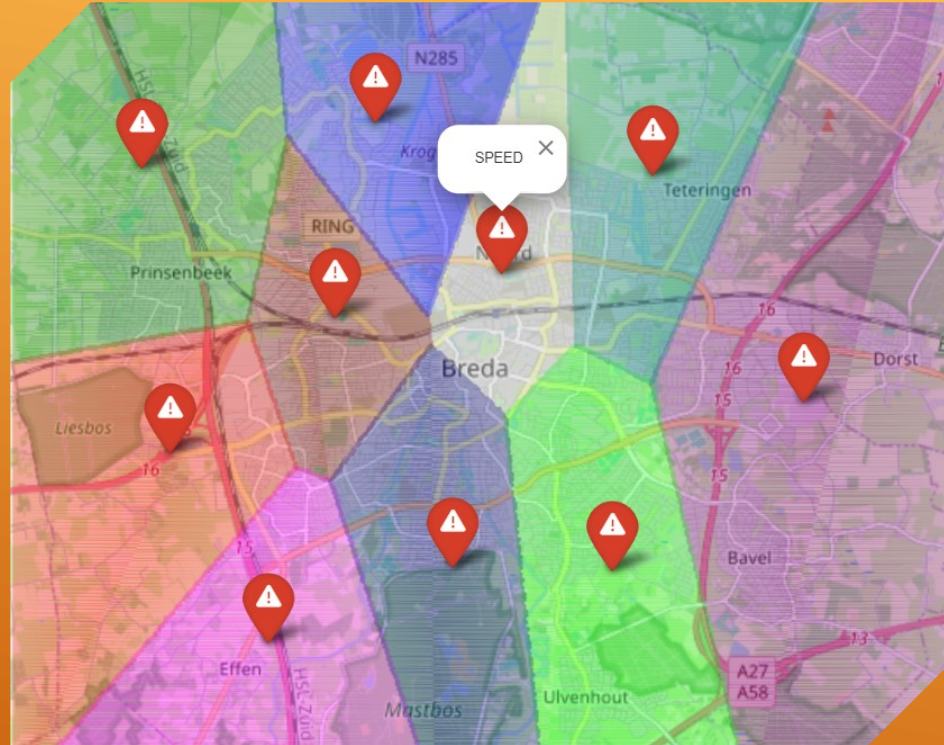
- ▢ Gradient Boosting
- ▢ 84% accuracy
- ▢ 9 Iterations
- ▢ Merged Dataset
- ▢ Hyperparameter Tuning
- ▢ Complex ML Model





Applications

- ▮ **Streamlit (Main Stakeholders)**
- ▮ **Flutterflow (Customers ANWB)**
- ▮ **99% Test Coverage**



References

Article 6: Classification Rules for High-Risk AI Systems | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/article/6/**](https://artificialintelligenceact.eu/article/6/)

Article 9: Risk Management System | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/article/9/**](https://artificialintelligenceact.eu/article/9/)

Article 10: Data and Data Governance | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/article/10/**](https://artificialintelligenceact.eu/article/10/)

Article 11: Technical Documentation | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/article/11/**](https://artificialintelligenceact.eu/article/11/)

Article 13: Transparency and provision of information to deployers | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/article/13/**](https://artificialintelligenceact.eu/article/13/)

Article 14: Human Oversight | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/article/14/**](https://artificialintelligenceact.eu/article/14/)

Article 15: Accuracy, Robustness and Cybersecurity | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/article/15/**](https://artificialintelligenceact.eu/article/15/)

Annex III: High-Risk AI Systems | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/annex/3/**](https://artificialintelligenceact.eu/annex/3/)

Annex IV: Technical Documentation | EU Artificial Intelligence Act. (n.d.). [**https://artificialintelligenceact.eu/annex/4/**](https://artificialintelligenceact.eu/annex/4/)